

Is an absolute level required?
Yes - an absolute number

No - relative is enough

PO integral + measurement anchor <i>Geometry sets shape, measurement sets level.</i> Absolute drone RCS in dBsm the uncertainty travels with it grid dither at $\lambda/16$: 1.78 dB PO validity floor: 0.729 λ Detection range and Pd benchmark σ sits directly in the radar equation size span a path solver misses: 64.1 dB Mesh fidelity budget for a drone target specular-only echo collapses with facets collapse over the facet sweep: 49.0 dB	Sionna RT alone, and it is exact <i>The engine prints the absolute number.</i> Free-space link budget, absolute power measured against Friis RT vs Friis: 6.3e-07 dB Specular reflection strength off a wall Fresnel coefficients and polarization RT / analytic: 0.9997
Our PO surface integral, pattern only <i>The target term is needed, but only its shape.</i> Aspect pattern shape vs azimuth kernel checked against analytic PO kernel vs analytic PO: 0.201 dB Airframe-to-airframe ranking by shape same material and area, different shape sphere vs plate: 31.29 dB Micro-Doppler modulation shape needs complex E per part	Sionna RT alone <i>The engine is the whole answer.</i> Chamber geometry, shadowing, occlusion finite mesh, exact hit test Multipath delay, floor-ghost timing $\tau = \text{path length} / c$ CFAR threshold from target-free noise no target in the cell

No - it stays in the answer

Yes - it divides out

Does the target term cancel as a constant?

The split is the radar equation's own: propagation is the engine's term, target scattering is a separate one.